

Pose-Based Fall Prediction with Cooperative Tree and Skeleton-Feature Fusion Models

Final Experimental Report

SURF 2026 Fall Prediction Project

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Abstract

This report evaluates the final version of a pose-based system that classifies video windows as Normal, Pre-fall, or Fall. A HistGradientBoosting classifier provides the confirmed state, while an ST-GCN–TCN skeleton–feature model provides an additional early-warning signal. A cooperative decision rule, a light hidden Markov model (HMM), a static-lying activity-of-daily-living (ADL) filter, and a Fall latch produce separate precision-oriented confirmed and recall-oriented early outputs.

The evaluation contains 4,714 labeled 15-frame windows from 117 camera sequences and 93 independent video or trial groups. Grouped five-fold cross-validation keeps related windows together and assigns the two UP-Fall camera views from the same trial to the same outer fold. The final confirmed output achieved **91.83% mean accuracy**, **85.71% mean macro F1**, and **95.06% mean Fall recall**. The early-warning output reduced mean accuracy to 89.60% and macro F1 to 83.73%, but increased mean Pre-fall recall from 65.96% to 81.40%. This gain lowered Pre-fall precision from 77.19% to 57.42%. Both final outputs detected 76 of 78 Fall sequences.

The supplied reproduction procedure passed every listed SHA-256 check, regenerated the final five-fold JSON byte for byte, and passed all 104 automated tests in an isolated environment. The evidence supports a reproducible research prototype with an explicit precision–recall trade-off; it does not constitute external or clinical validation.

1 Purpose and scope

Fall prediction is a temporal recognition problem: the system must distinguish ordinary motion, the transition toward a fall, and the fall state itself. Using a single output for all decisions can hide an important operating trade-off. A conservative confirmed state can reduce unnecessary alarms, whereas an earlier warning can provide more time to react but may increase false alerts.

This report asks:

How well does the supplied dual-branch pose system support precision-oriented confirmed states and recall-oriented early warnings for Normal, Pre-fall, and Fall under grouped five-fold cross-validation?

The analysis is limited to the supplied project, data, code, models, and result files. No external literature was retrieved. The report therefore does not claim state-of-the-art performance, clinical validity, statistical significance, or generalization to unseen real-world environments.

2 Data and labels

The processed dataset combines pose sequences stored under the UR Fall and UP-Fall project directories. It contains 117 camera sequences corresponding to 93 independent video or trial groups. Each training and evaluation sample is a 15-frame window, adjacent samples advance by three frames, and the window label is the state assigned to its final frame.

Table 1: Class distribution in the final evaluation set.

Class	Windows	Share
Normal	3,007	63.79%
Fall	1,287	27.30%
Pre-fall	420	8.91%
Total	4,714	100.00%

The small Pre-fall class is an important constraint. Its support is less than one third of the Fall support and about one seventh of the Normal support. Accuracy alone would therefore give an incomplete view. Macro F1 and class-specific Pre-fall precision and recall are primary metrics alongside Fall recall.

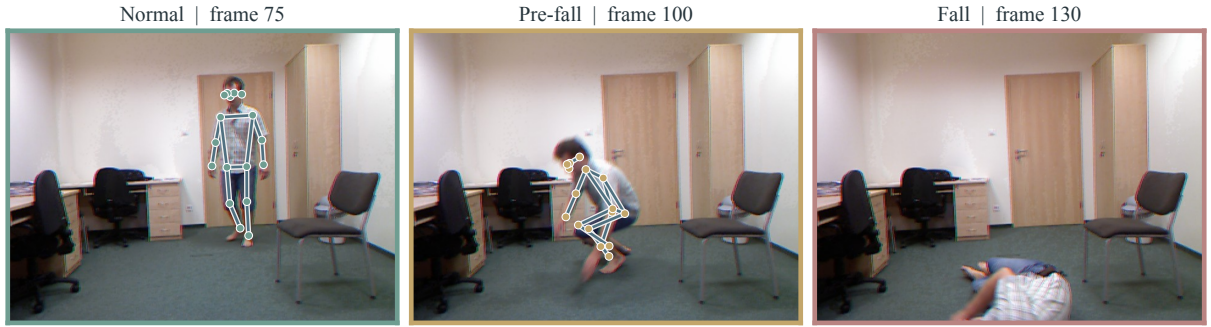


Figure 1: Three labeled frames from one supplied UR Fall sequence. The YOLO-derived skeleton is visible in the Normal and Pre-fall frames. In the Fall frame, the person partly exits the lower image boundary and the saved pose is unavailable. The example is qualitative and illustrates the need for temporal state handling; it is not a quantitative robustness test.

3 System method

3.1 Pose extraction and temporal windows

The runtime pose model is stored as yolo26n-pose.pt. It provides 17 COCO body keypoints. The saved landmark representation uses normalized coordinates and visibility values. The system processes a 15-frame temporal window with a stride of three frames.

3.2 Engineered-feature tree branch

The confirmed-state branch uses a HistGradientBoosting classifier. Its per-frame input contains 17 values: pose availability, torso angle, torso angular velocity, body-center position and change, vertical velocity,

body aspect ratio, body width and height, mean visibility, center drop, three feature-validity indicators, feature coverage, torso angular acceleration, and vertical acceleration.

This branch is authoritative in the cooperative policy. During full-model training, the project enables standing calibration, acceleration features, an eightfold Pre-fall class weight, and partial-pose augmentation.

3.3 Skeleton–feature fusion branch

The auxiliary branch receives two synchronized inputs:

- a skeleton tensor with shape $5 \times 15 \times 17$, whose channels are normalized x , normalized y , visibility, Δx , and Δy ;
- 13 engineered temporal features without the four additional robust-validity values used by the tree branch.

The skeleton encoder contains three spatial-temporal graph convolution blocks with 16, 32, and 32 channels. The feature encoder contains two causal temporal convolution blocks with 32 channels each. Each encoder combines its last, mean, and maximum temporal summaries before the embeddings are concatenated. A two-layer classifier produces three class logits. The full-data fusion model uses dropout 0.3, Pre-fall class weight 4.0, temperature 1.0826, and advisory threshold 0.655.

3.4 Cooperative decision and postprocessing

The decision policy is intentionally asymmetric:

1. A tree Fall is immediately a confirmed Fall.
2. A tree Pre-fall is a confirmed Pre-fall.
3. A fusion-only Pre-fall becomes an advisory while the confirmed state remains Normal.
4. A fusion-only Fall first becomes a Pre-fall advisory and becomes a Fall alert after three consecutive fusion decisions.
5. A light HMM smooths the calibrated fusion probabilities.
6. A static-lying ADL filter changes only a low, static posture without recent fall-like motion.
7. A confirmed dynamic or non-static Fall remains latched until acknowledgement.

The complete implementation is summarized in Figure 2. It is drawn from the supplied model artifacts and code and does not contain an LSTM.

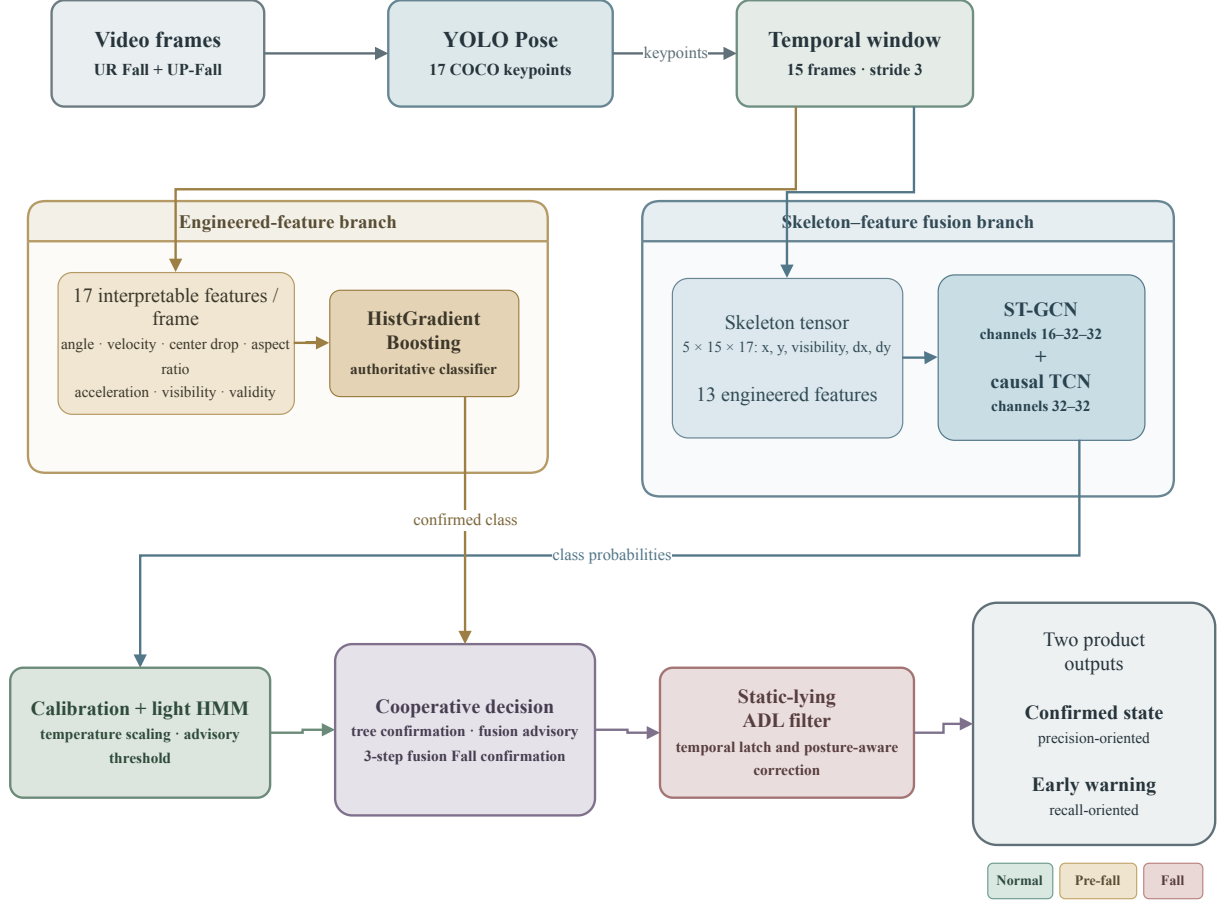


Figure 2: Final system architecture. The tree branch controls confirmed states, while the ST-GCN–TCN branch supplies calibrated, HMM-smoothed advisories. Cooperative decision and static-lying postprocessing produce separate confirmed and early-warning outputs. The figure is an editable Draw.io diagram exported with the official desktop engine.

4 Evaluation setup

The final report uses grouped five-fold cross-validation with random state 42. The outer split unit is a video or trial group rather than an individual window. Both cameras from the same UP-Fall trial stay in the same outer fold. The outer test set is not used to select the deep-model training epoch.

Outer-fold training sizes range from 3,745 to 3,795 windows, and test sizes range from 919 to 969 windows. The selected fusion-model training epochs are 45, 27, 16, 33, and 75 for folds 1–5. A separate nested procedure selects the Pre-fall class weight, temperature, and advisory threshold without using outer-test labels.

Two types of summary are reported:

- **five-fold mean $\pm$ sample standard deviation**, which describes variation across outer folds;
- **pooled out-of-fold metrics**, which concatenate predictions from all outer test folds and retain the exact total denominators.

The final postprocessing comparison is exploratory because the same outer-fold structure was previously inspected during component development. It should not be interpreted as a new untouched external validation.

5 Key results

5.1 Window-level performance

Table 2: Five-fold window-level performance, reported as mean $\pm$ sample standard deviation. PF denotes Pre-fall.

Output	Accuracy	Macro F1	PF precision	PF recall	Fall recall
Tree classifier	90.32 $\pm$ 1.77	84.19 $\pm$ 2.71	76.95 $\pm$ 1.55	66.67 $\pm$ 8.36	83.62 $\pm$ 6.16
Fusion + HMM	88.59 $\pm$ 2.65	83.12 $\pm$ 2.96	59.34 $\pm$ 3.14	85.23 $\pm$ 8.47	82.82 $\pm$ 5.05
Final confirmed	91.83 $\pm$ 2.59	85.71 $\pm$ 2.34	77.19 $\pm$ 1.59	65.96 $\pm$ 7.96	95.06 $\pm$ 3.37
Final early warning	89.60 $\pm$ 2.68	83.73 $\pm$ 2.21	57.42 $\pm$ 2.75	81.40 $\pm$ 8.54	95.06 $\pm$ 3.37

All values in Table 2 are percentages.

The confirmed output has the best mean accuracy and macro F1 among the listed outputs. Its main change relative to the tree-only result is higher Fall recall after temporal decision and postprocessing. The early-warning output increases Pre-fall recall by 15.44 percentage points relative to the confirmed output, but lowers Pre-fall precision by 19.77 points. Accuracy and macro F1 also decrease by 2.23 and 1.97 points, respectively.

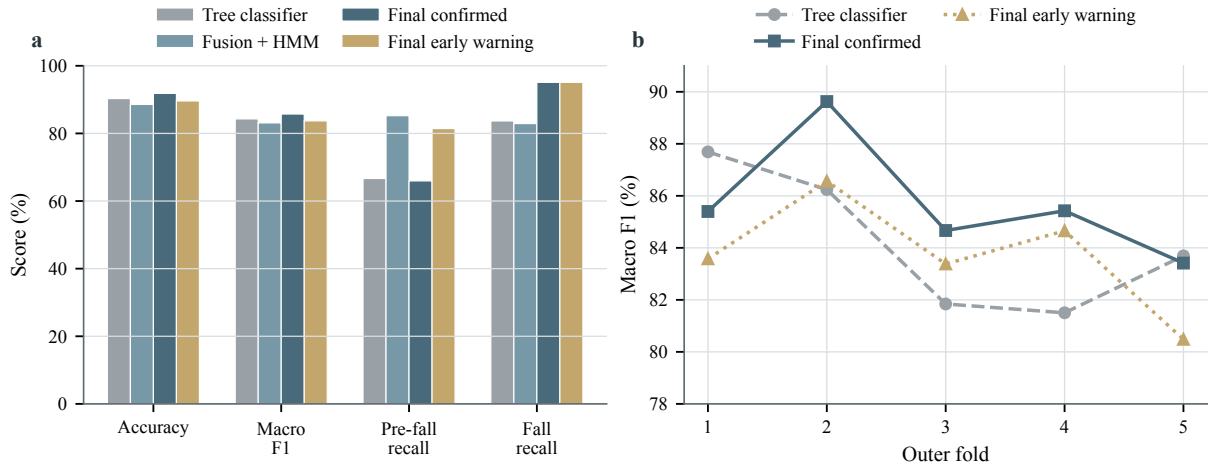


Figure 3: Window-level performance and fold stability. (a) The final confirmed output gives the strongest overall accuracy and macro F1, whereas the early-warning output trades precision for higher Pre-fall recall. The bar-chart axis starts at zero. (b) Macro F1 varies across outer folds; the line-chart axis is restricted to 78–91% to show fold variation.

The final confirmed macro F1 ranges from 83.41% to 89.63% across folds. This variation does not change the main interpretation: the confirmed output remains the stronger general-purpose summary, and the early output remains a higher-recall advisory.

5.2 Error distribution

The pooled out-of-fold confirmed confusion matrix is

$$\begin{bmatrix} 1224 & 37 & 26 \\ 123 & 2828 & 56 \\ 47 & 96 & 277 \end{bmatrix},$$

where rows and columns follow Fall, Normal, and Pre-fall. It corresponds to 95.10% Fall recall, 94.05% Normal recall, and 65.95% Pre-fall recall.

For the early-warning output, the pooled matrix is

$$\begin{bmatrix} 1224 & 23 & 40 \\ 133 & 2658 & 216 \\ 47 & 31 & 342 \end{bmatrix}.$$

Pre-fall recall increases to 81.43%, mainly because fewer true Pre-fall windows are mapped to Normal. At the same time, 216 Normal windows are mapped to Pre-fall instead of 56 for the confirmed output. This is the principal source of lower Pre-fall precision.

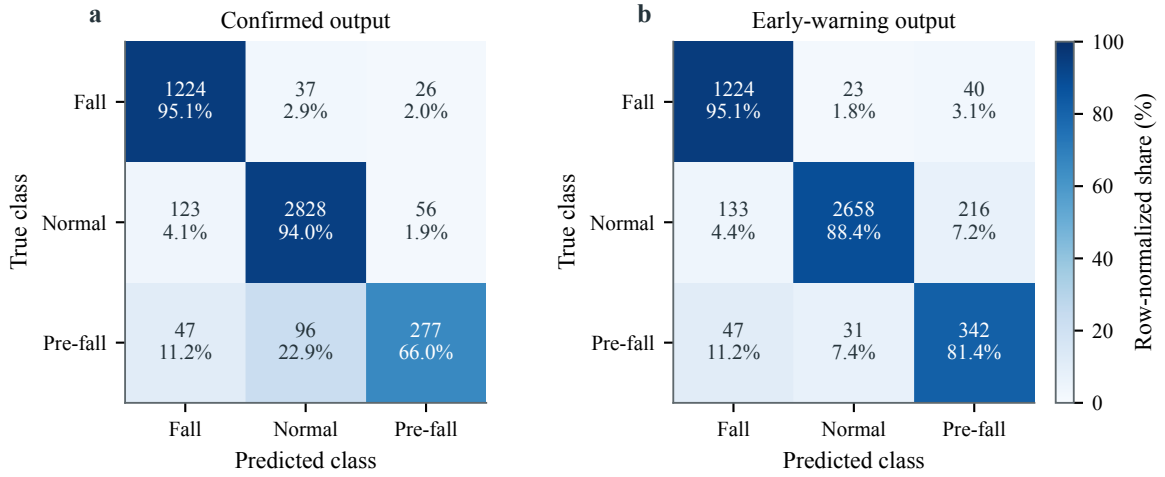


Figure 4: Pooled out-of-fold confusion matrices. The early-warning output retains the same number of correctly detected Fall windows and increases correct Pre-fall windows from 277 to 342, but it also increases Normal-to-Pre-fall warnings from 56 to 216. Each cell gives the count and row-normalized percentage.

5.3 Sequence-level performance

At sequence level, the final confirmed and early-warning outputs each detect 76 of 78 Fall sequences, a detection rate of 97.44%. The fusion + HMM component detects 77 Fall sequences, but it produces some form of warning in 20 of 39 non-Fall sequences. The final confirmed output reduces this warning burden to 9 of 39 non-Fall sequences.

Table 3: Sequence-level outcomes.

Output	Fall detected	Non-Fall: any warning	Non-Fall: false Fall
Tree classifier	76 / 78	12 / 39	8 / 39
Fusion + HMM	77 / 78	20 / 39	5 / 39
Final confirmed	76 / 78	9 / 39	5 / 39
Final early warning	76 / 78	24 / 39	6 / 39

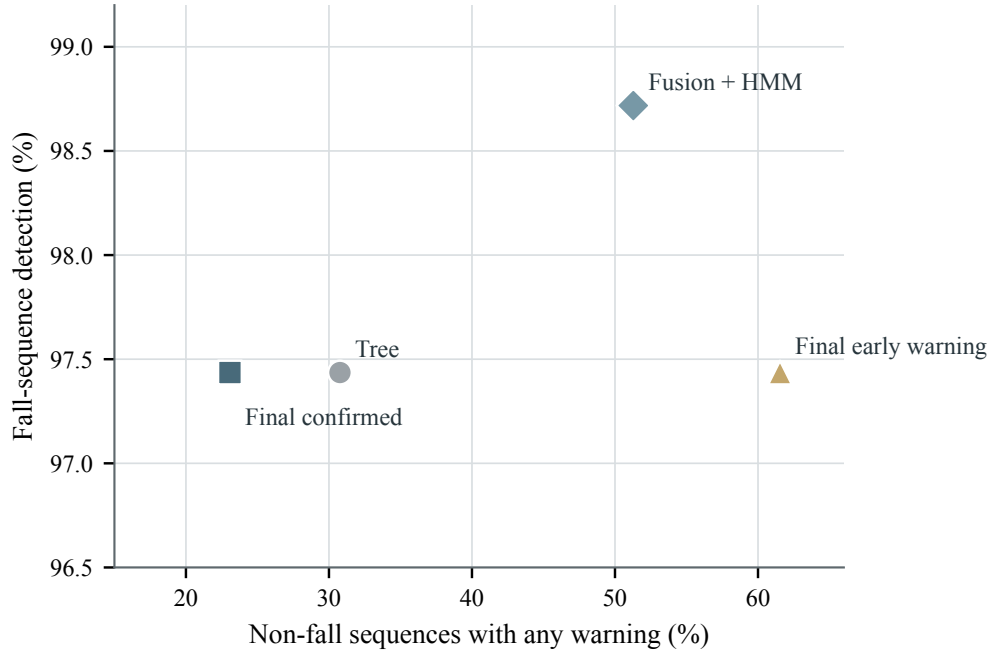


Figure 5: Sequence-level operating trade-off. The fusion + HMM component detects one additional Fall sequence but warns on more than half of the non-Fall sequences. The final confirmed output has the lowest non-Fall warning rate among the compared outputs while retaining 76 of 78 Fall detections.

The sequence result reinforces the intended product separation. A low-level advisory can be made sensitive, but it should not be presented as an equally reliable confirmed state.

6 Diagnosis and interpretation

Three findings are most important.

The fusion branch is a sensitivity-oriented signal, not a consistent replacement for the tree branch. Its mean Pre-fall recall is high, but its accuracy, macro F1, and Pre-fall precision are lower than the tree values. The final design therefore uses it as an advisory rather than as sole authority.

Temporal decision and postprocessing substantially increase Fall recall in the confirmed output. In pooled out-of-fold results, Fall recall rises from 83.68% for the tree result to 95.10% for the final confirmed output. Normal recall decreases from 96.44% to 94.05%. This is a controlled shift toward Fall sensitivity rather than a uniform improvement in every class.

Pre-fall remains the hardest state. It is the smallest class, lies between ordinary activity and Fall, and depends on a frame-level transition boundary. The early-warning threshold recovers more Pre-fall

windows, but many Normal windows also cross the advisory threshold. Future improvement should focus on boundary quality and false-warning analysis rather than on sensitivity alone.

7 Limitations

1. **No untouched external test set.** The final comparison uses the same outer-fold structure inspected during component development. The reported differences are exploratory.
2. **Limited data diversity.** The evaluation covers the supplied UR Fall and UP-Fall project data. It does not establish performance in homes, hospitals, different camera placements, or different populations.
3. **Window overlap.** Windows overlap because the stride is three frames. Grouped splitting prevents the same video or trial from crossing folds, but neighboring test windows remain correlated.
4. **Pre-fall label uncertainty.** A window receives the label of its final frame. Results may change with another transition boundary or annotation policy.
5. **Exploratory postprocessing.** The static-lying and cooperative rules were evaluated on previously inspected fold outputs.
6. **No latency measurement.** The supplied materials do not include measured frame rate, end-to-end delay, memory use, or hardware benchmarks.
7. **No uncertainty or significance analysis.** Five-fold standard deviations are reported, but repeated cross-validation, confidence intervals, and statistical tests are unavailable.
8. **Pose loss remains possible.** The qualitative sequence shows that a person may leave the image boundary during a Fall. Temporal state handling helps preserve system behavior, but one example does not quantify robustness.
9. **Deployment governance is outside scope.** Dataset licensing, consent, privacy safeguards, alarm escalation, and clinical risk controls are not documented in the supplied corpus.

8 Reproducibility status

Final verification used a project-local Python 3.11 environment. Recorded package versions for result reproduction and tests are NumPy 2.4.6, SciPy 1.17.1, scikit-learn 1.9.0, joblib 1.5.3, PyTorch 2.12.1, Matplotlib 3.11.0, OpenCV 4.13.0.92, and pytest 9.1.1.

The supplied reproduction script completed three checks:

1. verified all listed final model, annotation, and report SHA-256 values;
2. loaded the five saved fusion fold models and regenerated the complete out-of-fold result;
3. matched the regenerated JSON against the final reference report byte for byte.

The first isolated test run passed 103 tests and failed one import because the minimal environment did not yet include OpenCV. After `opencv-python==4.13.0.92` was added, the full suite passed:

104 passed, 1 warning in 42.05 s.

The warning reported that joblib could not determine the number of physical cores and therefore used the logical-core count. It did not change the exit status or test results.

The verified scope is result-file reproduction and the current automated test suite. This report does not claim that every model was retrained from raw video frames during verification.

9 Final conclusions and recommended use

The current system should expose both outputs:

- Use **Final confirmed** as the official Normal / Pre-fall / Fall state. It provides the stronger overall balance, with 91.83% mean accuracy, 85.71% mean macro F1, and 95.06% mean Fall recall.
- Use **Final early warning** as a separate advisory channel when higher Pre-fall sensitivity is useful. It raises mean Pre-fall recall to 81.40%, but its 57.42% precision means that the advisory should not be presented as a confirmed state.

For the next validation stage, the smallest high-value task is to freeze all thresholds and postprocessing rules, then evaluate once on a genuinely untouched subject, site, or dataset. That evaluation should include event-level false-warning duration, detection lead time, missed-Fall review, and measured end-to-end latency. Until that evidence is available, this report supports a reproducible research prototype and an explicit operating trade-off, not a deployment-ready safety system.